

English

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CLASSROOM CONTACT PROGRAMME
(Academic Session : 2024-2025)

Telegram: @NEETxNOAH

Test Pattern
NEET (UG)
MAJOR
01-04-2025

PRE-MEDICAL : ENTHUSIAST COURSE (RBT)

Test Booklet Code

This Booklet contains **28** pages.

E1

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

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Centre of Examination (in Capitals) : _____

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Your Target is to secure Good Rank in Pre-Medical 2025

SUBJECT : PHYSICS

Topic : NLM, FRICTION, WPE, CIRCULAR MOTION

1. The coefficient of friction (μ) and angle of friction (λ) are related as :

(1) $\sin \lambda = \mu$ (2) $\cos \lambda = \mu$
(3) $\tan \lambda = \mu$ (4) $\sec \lambda = \mu$

2. A ball of mass 250 gm collides on the wall of billiards table with 2 m/s. After remaining in contact with the surface for 0.02 s, it rebounds with 1.8 m/s. Calculate approximately the magnitude of average force exerted on ball by the surface.

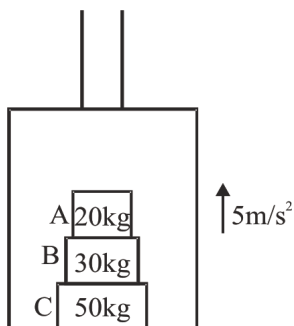
(1) 40 N (2) 48 N
(3) 25 N (4) None

3. **Assertion (A)** :- Magnitude of the contact force can never be greater than the magnitude of frictional force.

Reason (R) :- Contact force is the resultant of the friction force and normal reaction.

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is incorrect but (R) is correct
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

4. In shown situation, elevator (lift) is moving upward with acceleration of 5 m/s^2 . Find net force on B?

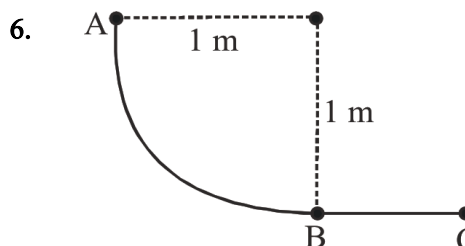


(1) 150 N (2) 200 N
(3) 250 N (4) 100 N

5. **Assertion** : The acceleration of a body down a rough inclined plane is greater than the acceleration due to gravity.

Reason : The body is able to slide on a inclined plane only when its acceleration is greater than acceleration due to gravity.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are false.



In the track shown in figure section AB is a quadrant of a circle of 1m radius. A block is released from point A and slides without friction until it reaches point B. After point B it moves on a rough horizontal floor and comes to rest after travelling the distance 3 m from point B. What is the coefficient of friction between floor and block?

(1) $1/3$ (2) $2/3$ (3) $1/4$ (4) $3/8$

7. **Assertion** : No work is done if the displacement is perpendicular to the direction of the applied force.

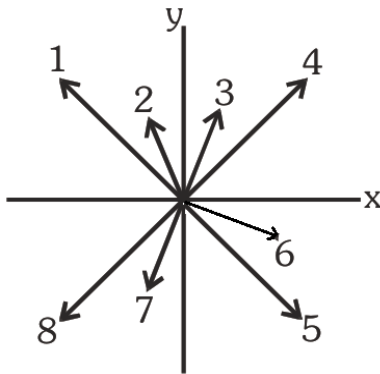
Reason : If the angle between the force and displacement vector is obtuse, the W.D is negative.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

8. A body of mass 0.01 kg starts to move in a viscous liquid with initial velocity 5 m/s. If drag force is 10^{-3} N then find time after which velocity will become $\frac{1}{10}$ m/s

(1) 49 sec (2) 100 sec
(3) 149 sec (4) 200 sec

9. Two forces $\vec{F}_1 = (3\text{N})\hat{i} - (4\text{N})\hat{j}$ and $\vec{F}_2 = -(1\text{N})\hat{i} - (2\text{N})\hat{j}$ act on a point object. In the given figure which of the eight vectors represents $\vec{F}_1$ and $\vec{F}_2$? What is the magnitude of the net forces?

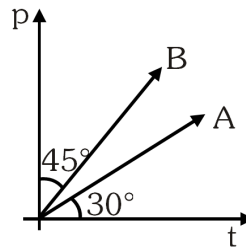


- (1) 4 and 8; $\sqrt{19}$ N
(2) 3 and 6; 5 N
(3) 5 and 7; $\sqrt{40}$ N
(4) 2 and 6; 7 N
10. A box is lying on an inclined plane. If the box starts sliding when the angle of inclination is 60° , then the coefficient of static friction of the box and plane is-
- (1) 2.732 (2) 1.732
(3) 0.267 (4) 0.176
11. When a force of constant magnitude and a fixed direction acts on a moving object, then its path is:
- (1) Circular (2) Parabolic
(3) Straight line (4) Either (2) or (3)

12. A particle is acted upon by a force given by $F = (12t + 3t^2)\text{N}$, where t is in seconds. Find the change in momentum of that particle from $t = 1$ to $t = 3$ sec.

(1) 74 kg m/s² (2) 89 kg-m/s²
(3) 737 kg-m/s² (4) None of the above

13. For the given momentum-time (p-t) graphs find the ratio of force on particles A & B.



(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{2}$

14. When a same force is applied on two different objects. It produces accelerations of 4 m/sec² and 6 m/sec². If the same force is applied on the combination of these objects, calculate acceleration of combination.

(1) 10 m/s² (2) 2.4 m/s²
(3) $\sqrt{24}$ m/s² (4) 4.8 m/s²

15. The exhaust velocity of gases with respect to a small rocket of mass 25 kg is 28×10^2 m/s. At what rate the fuel must burn so that it may rise up with an acceleration of 9.8 m/s² in earth gravitation field?

(1) 175 Kg (2) 1.75 Kg/s
(3) 0.175 Kg/s (4) Zero

16. When a resultant external force acts on moving body

(a) both momentum ($\vec{p}$) and K.E. may change
(b) neither momentum ($\vec{p}$) nor K.E. changes
(c) momentum ($\vec{p}$) may change but not the K.E.
(d) K.E. may change but not the momentum ($\vec{p}$)

(1) a, b (2) a, d (3) b, c (4) a, c

17. **Assertion :** The work done by a non-conservative force is always negative.

Reason : When a non-conservative force is involved in a system, it always dissipates energy.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

18. A particle is moving in a potential region given by $U = K(x^2 + y^2 + z^2)$. The force acting on the particle is given by :-

- (1) $-2K(\hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k})$
- (2) $K(\hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k})$
- (3) $\frac{K}{2}(\hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k})$
- (4) $K(x^2\hat{i} + y^2\hat{j} + z^2\hat{k})$

19. Which statements are not correct :

- (a) A body can have momentum without K.E.
- (b) The relative velocity of the two particles in head on collision is unchanged.
- (c) P.E of an object can have a negative value.
- (d) K.E. of an object can have a negative value.

- (1) a, c, d
- (2) a, b, c
- (3) a, b, d
- (4) a, b, c, d

20. Co-efficient of restitution for inelastic collision is -

- (1) $e = 1$
- (2) $e = 0$
- (3) $0 \leq e < 1$
- (4) $e > 1$

21. **Assertion :** The change in K.E. of a particle is equal to the work done on it by the net force.

Reason : Change in K.E. of particle is equal to the work done only in case of a system of one particle.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

22. The potential energy of a particle varies with position x according to the relation

$$U(x) = 2x^4 - 27x, \text{ the point } x = \frac{3}{2} \text{ is point of -}$$

- (1) Unstable equilibrium
- (2) Stable equilibrium
- (3) Neutral equilibrium
- (4) None of these

23. **Statement-1 :** Graph between potential energy of a spring versus the extension or compression of the spring is a parabola.

Statement-2 : Potential energy of a stretched or compressed spring is proportional to square of extension or compression.

- (1) Statement-1 is True; Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (2) Statement-1 is True; Statement-2 is True; Statement-2 is NOT explanation for Statement-1
- (3) Statement-1 is True, Statement-2 is False
- (4) Statement-1 is False, Statement-2 is True

24. A small block of mass m is kept on a rough inclined surface of inclination θ fixed in an elevator. The elevator goes down with a uniform velocity v and the block does not slide on the wedge. The work done by the force of friction on the block in time t will be :-

(1) zero (2) $-mgvt \cos^2 \theta$
(3) $-mgvt \sin^2 \theta$ (4) $mgvt \sin^2 \theta$

25. If the momentum of a body is increased by 200 % then increase in its kinetic energy will be :-

(1) 200% (2) 600% (3) 800% (4) 100%

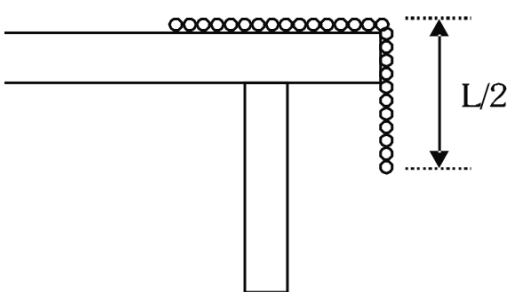
26. A 0.5 kg ball is thrown up with an initial speed 14 m/s and reaches a maximum height of 9 m. How much work is done by air drag acting on the ball during the ascent ($g = 9.8 \text{ m/s}^2$)?

(1) -19.6 J (2) -4.9 J
(3) -10 J (4) -9.8 J

27. Under the action of force, 3 kg body moves such that $x = \frac{t^2}{2}$ where position x is in meter and t is in second. The work done by the force in first 3 second is :-

(1) 13.5 J (2) 27 J (3) 81 J (4) 109 J

28. A uniform chain (mass M , length L) is released from rest from a smooth horizontal surface as shown in the figure. Velocity of the chain at the instant it completely comes out of the table will be.

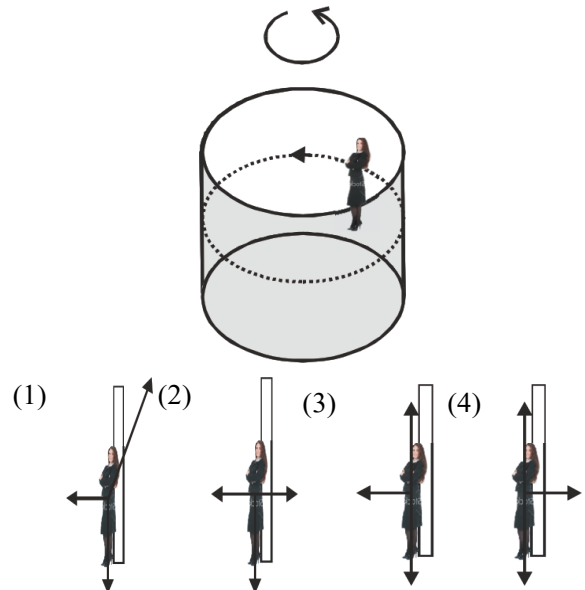


(1) $v = \sqrt{gL}$ (2) $v = \sqrt{\frac{3gL}{4}}$
(3) $v = \frac{1}{2} \sqrt{5gL}$ (4) $v = \frac{1}{2} \sqrt{gL}$

29. A force $\vec{F} = 6\hat{i} + 2\hat{j} - 3\hat{k}$ acts on a particle and produces a displacement of $\vec{s} = 2\hat{i} - 3\hat{j} - x\hat{k}$. If the work done is zero, the value of x is :-

(1) -2 (2) 1/2 (3) 6 (4) 5

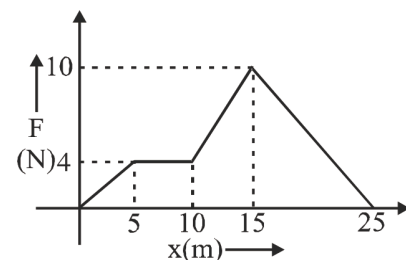
30. A girl finds herself stuck with her back to the wall of a cylinder rotating about its axis. Which diagram correctly shows the forces acting on her? (with respect to ground)



31. A block of mass 50 kg slides over a horizontal distance of 1m. If the coefficient of friction between their surface is 0.2, then work done against friction is :-

(1) 98 J (2) 72 J (3) 56 J (4) 34 J

32. The variation of force acting on a particle along the x -axis is shown in the figure. Find the work done by force during displacement $x = 0$ to $x = 25 \text{ m}$.



(1) 100 J (2) 115 J
(3) 95 J (4) 80 J

33. A man M_1 of mass 80 kg runs up a staircase in 15 s. Another man M_2 , also of mass 80 kg, runs up the same staircase in 20 s. The ratio of the power developed by them will be-

- (1) 1 (2) $4/3$
(3) $16/9$ (4) None of the above

34. A ball is thrown vertically upwards with a velocity v and an initial kinetic energy E . When half way to the top of its flight, it can have velocity and kinetic energy respectively of :

- (1) $\frac{v}{2}, \frac{E}{2}$ (2) $\frac{v}{\sqrt{2}}, \frac{E}{2}$
(3) $\frac{v}{4}, \frac{E}{2}$ (4) $\frac{v}{4}, \frac{E}{\sqrt{2}}$

35. A car is moving in a circular horizontal track of radius 10 m with a constant speed of 10 m/sec. A bob is suspended from the roof of the car by a light rigid rod of length 1.00 m. The angle made by the rod with the track is :-

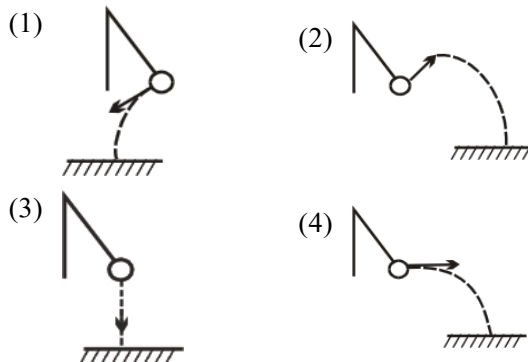
- (1) zero (2) 30°
(3) 45° (4) 60°

36. In vertical circular motion of a bob, match the entries of list-I with entries of list-II. Here, v_0 is the velocity of bob at lowest point.

	List-I (speed at lowest point)		List-II (possible situation)
(P)	$v_0 = \sqrt{5g\ell}$	(1)	Tension at lowest point = $6mg$
(Q)	$v_0 = \sqrt{g\ell}$	(2)	string will slack for a finite time
(R)	$v_0 = 2\sqrt{g\ell}$	(3)	bob will oscillate
(S)	$v_0 = 3\sqrt{g\ell}$	(4)	Tension at highest point = $4mg$

- (1) P-1, Q-2, R-3, S-4 (2) P-4, Q-2, R-3, S-1
(3) P-1, Q-3, R-2, S-4 (4) P-4, Q-3, R-2, S-1

37. A pendulum bob is swinging in a vertical plane such that its angular amplitude is less than 90° . At its highest point, the string is cut. Which trajectory is possible for the bob afterwards ?



38. A flywheel rotates with a uniform angular acceleration. Its angular velocity increases from 20π rad/s to 40π rad/s in 10 sec. How many rotations did it make in this period :-

- (1) 50 (2) 150
(3) 100 (4) 250

39. A particle is moving around a circular path with uniform angular speed (ω). The radius of the circular path is (r). The acceleration of the particle is :-

- (1) $\frac{\omega^2}{r}$ (2) $\frac{\omega}{r}$
(3) $v\omega$ (4) vr

40. If a car is to travel with a speed v along the frictionless, banked circular track of radius r , the required angle of banking so that the car safely negotiates the curve is :-

- (1) $\theta = \tan^{-1} \left(\frac{v^2}{rg} \right)$
(2) $\theta = \tan^{-1} \left(\frac{v}{rg} \right)$
(3) $\theta = \tan^{-1} \left(\frac{r^2}{vg} \right)$
(4) $\theta < \tan^{-1} \left(\frac{v}{rg} \right)$

41. Match the following :-

	Column-I		Column-II
(A)	Work	(p)	Slope of momentum-time graph
(B)	Power	(q)	Slope of work-time graph
(C)	Force	(r)	Area under force-position graph
(D)	Impulse	(s)	Area under force-time graph

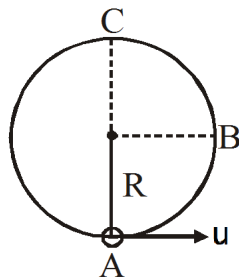
(1) A-q ; B-p ; C-s ; D-r

(2) A-r ; B-q ; C-p ; D-s

(3) A-s ; B-q ; C-p ; D-r

(4) A-q ; B-r ; C-s ; D-p

42. A particle is suspended from a string of length R . It is given a velocity $u = 3\sqrt{gR}$ at the bottom. Find velocity at B:-



(1) $\sqrt{7gR}$

(2) $\sqrt{5gR}$

(3) $\sqrt{3gR}$

(4) $\sqrt{8gR}$

43. A particle moves in a circular path of radius R with an angular velocity $\omega = a - bt$ where a and b are positive constants and t is time. The magnitude of the acceleration of the particle at time $\frac{2a}{b}$ is :-

(1) $\frac{a}{R}$

(2) a^2R

(3) $R(a^2 + b^2)$

(4) $R\sqrt{a^4 + b^2}$

44. A particle is moving in a vertical circle. The tension in the string when passing through two positions at angle of 30° and 60° from vertical (the lowest position) are T_1 and T_2 respectively, then:-

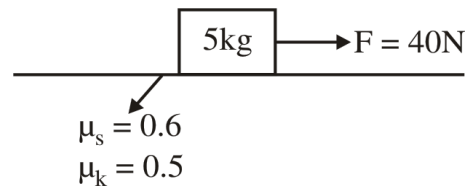
(1) $T_1 = T_2$

(2) $T_2 > T_1$

(3) $T_1 > T_2$

(4) Tension in the string always remains the same.

45.



Match the columns :

Column-I		Column-II	
(P)	$a(\text{m/s}^2)$	(1)	25
(Q)	N (newton)	(2)	30
(R)	f_L (newton)	(3)	50
(S)	f_K (newton)	(4)	3

(1) $P \rightarrow 3, Q \rightarrow 4, R \rightarrow 2, S \rightarrow 1$

(2) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 1$

(3) $P \rightarrow 1, Q \rightarrow 2, R \rightarrow 3, S \rightarrow 4$

(4) $P \rightarrow 4, Q \rightarrow 2, R \rightarrow 1, S \rightarrow 3$

SUBJECT : CHEMISTRY

Topic : CHEMICAL BONDING, THERMODYNAMIC

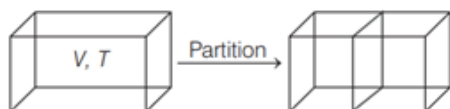
46. Which among the following is diamagnetic molecule and follow octet rule—
 (1) NO_2 (2) ClO_2
 (3) CO_2 (4) NO
47. Which of the following have one lone pair on the central atom ?
 (I) NH_3
 (II) SF_4
 (III) XeO_4
 (IV) XeF_6
 (1) I, II (2) I, II, IV
 (3) I, IV (4) I, II, III, IV
48. Which amongst the following is not isoelectronic :-
 (1) NO (2) CN^{-1}
 (3) N_2 (4) O_2^{+2}
49. Correct order of radius ?
 (1) $\text{F} > \text{F}^- > \text{F}^+$
 (2) $\text{F}^- > \text{Cl}^- > \text{Br}^-$
 (3) $\text{Cl}^- > \text{Br}^- > \text{H}^+ > \text{Li}^{2+}$
 (4) $\text{Te}^{2-} > \text{I}^- > \text{Br}^- > \text{Na}^+$
50. Select the compound which is thermally unstable :-
 (1) Li_2CO_3
 (2) Na_2CO_3
 (3) K_2CO_3
 (4) BeCO_3
51. Which among the following does not forms a infinite three dimensional giant structure ?
 (1) AlN (2) SiO_2
 (3) CO_2 (4) $\text{C}_{(\text{diamond})}$
52. Which of the following is ionic compound
 (1) AlCl_3 (2) AlF_3
 (3) LiCl (4) BeCl_2
53. In which of the following species is polar ?
 (a) C_6H_6 (b) XeF_2 (c) SO_2 (d) SF_6
 (e) SF_4
 Correct answer is :-
 (1) b and d (2) a, b and e
 (3) a and e (4) c and e
54. Boiling point is highest for :-
 (1) CH_4 (2) NaCl
 (3) CCl_4 (4) CHCl_3
55. Bond order in N_2^+ is :-
 (1) $\frac{6}{2}$ (2) $\frac{4}{2}$
 (3) $\frac{5}{2}$ (4) $\frac{3}{2}$
56. The increasing order of reactivity of halogens is :-
 (1) $\text{I}_2 < \text{Br}_2 < \text{Cl}_2 < \text{F}_2$
 (2) $\text{Cl}_2 < \text{F}_2 < \text{Br}_2 < \text{I}_2$
 (3) $\text{Cl}_2 < \text{Br}_2 < \text{I}_2 < \text{F}_2$
 (4) $\text{I}_2 < \text{Cl}_2 < \text{Br}_2 < \text{F}_2$
57. Which of the following alkaline earth metal is not easily converted into its carbonate when it is kept in moist air :-
 (1) Mg (2) Ca
 (3) Sr (4) Be
58. Which of the following is an examples of super octet molecule (expansion of octet) ?
 (1) ClF_3 (2) PCl_5
 (3) IF_7 (4) All the three

59. An element A has configuration $1s^2 2s^2 2p^6 3s^2$ formula of its phosphate is :-
 (1) ASO_4 (2) APo_4
 (3) $A_3(PO_4)_2$ (4) $A_2(PO_4)_3$
60. Which is odd electron species :-
 (1) CO_2 (2) NO_2
 (3) SO_2 (4) SiO_2
61. When lewis Acid and lewis base are present then bond between them will be :-
 (1) Ionic
 (2) Covalent
 (3) Coordinate
 (4) Metallic
62. The hybridization of O in H_3O^+ is:
 (1) sp (2) sp^2
 (3) sp^3d (4) sp^3
63. An ionic compound $A^+ B^-$ is most likely to be formed when –
 (1) Ionization energy of A is low
 (2) Electron affinity of B is high
 (3) Electron affinity of B is low
 (4) Both (1) and (2)
64. Choose the compound from each of the following pairs that has the more solubility in water :-
 (a) $BeSO_4$, $BaSO_4$
 (b) $LiOH$, $CsOH$
 (c) $AgCl$, AgI
 (1) $BeSO_4$, $CsOH$, AgI
 (2) $BeSO_4$, $CsOH$, $AgCl$
 (3) $BaSO_4$, $LiOH$, $AgCl$
 (4) $BaSO_4$, $LiOH$, AgI
65. The cause of chemical combination is :-
 (1) Tendency of acquire minimum energy
 (2) Tendency to acquire noble gas configuration
 (3) (1) & (2) both
 (4) None of these
66. A π bond is formed by the overlap of
 (1) s-s orbitals
 (2) p-p orbitals in coaxial manner
 (3) s-p orbitals
 (4) p-p orbitals in collateral manner
67. Due to inter molecular H-bond generally :-
 (1) Solubility increases
 (2) Boiling point increases
 (3) Viscosity increases
 (4) All of these
68. In an adiabatic closed system, an ideal gas is allowed to expand both reversibly and irreversibly. If the initial temperature is T_i and the final temperature is T_f , then which of the following statement is correct?
 (1) $T_f > T_i$ (reversible) but $T_f = T_i$ (irreversible)
 (2) $(T_f)_{irrev} > (T_f)_{rev}$
 (3) $(T_f)_{irrev} = (T_f)_{rev}$
 (4) $T_f = T_i$ for both reversible and irreversible process
69. A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0mL to 375mL at a constant temperature of $37.0^\circ C$. As it does so, it absorbs 208J of heat. The value of q and w for the process will be:
 ($R=8.314J/molK$)($\ln 7.5=2.01$)
 (1) $q=+208J, w=-208J$ (2) $q=+208J, w=-360J$
 (3) $q=-208J, w=+360J$ (4) $q=+208J, w=+208J$

70. When 1 mole of ice melts at 0°C and at a constant pressure of 1 atm, 1440 cal of heat is absorbed by the system. If the molar volume of ice and water are 0.0196 and 0.0180 L, then the value of ΔH and ΔU will be

- (1) 1440 cal, 1440.039 cal (2) 1440 cal, 1438 cal
(3) 1440.039 cal, 1440 cal (4) 1438 cal, 1440 cal

71. The correct inference that can be observed from the given figure is



- (1) Each part has double the temperature of the gas
(2) Each part has the same temperature of the gas
(3) Each part only have half the temperature of the gas
(4) Each part has one-fourth the temperature of the gas

72. **Assertion (A):** The difference between ΔH and ΔU is not significant for systems consisting of only solids or liquids.

Reason (R): Solids and liquids do not suffer any significant volume changes upon heating.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

73. Enthalpy of sublimation of iodine is 24 cal g^{-1} at 200°C . If specific heat of $\text{I}_2(\text{s})$ and $\text{I}_2(\text{vap})$ are 0.055 and $0.031 \text{ cal g}^{-1}\text{K}^{-1}$ respectively, then enthalpy of sublimation of iodine at 250°C in cal g^{-1} is :

- (1) 2.85 (2) 5.7 (3) 22.8 (4) 11.4

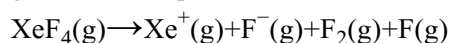
74. Standard enthalpy of vaporisation $\Delta V_{\text{vap}}.H^{\circ}$ for water at 100°C is $40.66 \text{ kJ mol}^{-1}$. The internal energy change of Vaporization of water at 100°C (in kJ mol^{-1}) is (Assume water vapour to behave like an ideal gas).

- (1) + 37.56 (2) - 43.76
(3) +43.76 (4) +40.66

75. The standard enthalpy of formation of $\text{NH}_3(\text{g})$ is $-81.9 \text{ kJ mol}^{-1}$. The amount of heat required to decompose 26 g of $\text{NH}_3(\text{g})$ into its elements is

- (1) 125.26 (2) 143.8
(3) 164.7 (4) 105.6

76. Calculate the bond enthalpy of Xe -F bond as given in the equation



[Given $\Delta_r H = 292 \text{ K cal mol}^{-1}$

Ionisation energy of Xe = 279 K cal / mol

Bond energy (F-F) = 38 Kcal/mol

Electron affinity of F = 85 Kcal/mol]

- (1) 8.5 kcal/mol (2) 34 kcal/mol
(3) 24 kcal/mol (4) 2.5 kcal/mol

77. Select the correct statement from the following.

I. The greater the disorder in an isolated system, the lower is the entropy.

II. The entropy change can be attributed to rearrangement of atoms or ions from one pattern in the reactants to another (in the product)

III. Decrease of regularity in structure would mean increase in entropy.

IV. The gaseous state is the state of highest entropy.

V. In a bomb calorimeter, heat absorbed at constant pressure can be measured.

Choose the correct answer from the options given below.

- (1) II, III and V (2) I and V
(3) I, II and III (4) II, III and IV

78. The enthalpy of solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$ and $\text{CuSO}_4(\text{s})$ are respectively -11.7 and -65.5 kJ/mol . The hydration enthalpy of anhydrous CuSO_4 will be

- (1) 53.8 kJ mol^{-1} (2) -9.8 kJ mol^{-1}
 (3) $-77.2 \text{ kJ mol}^{-1}$ (4) $-53.8 \text{ kJ mol}^{-1}$

79. Which of the following pair(s) of conditions are certain to result in a spontaneous reaction?

- (1) Endothermic and decreasing order
 (2) Exothermic and increasing disorder
 (3) Endothermic and increasing disorder
 (4) Exothermic and decreasing disorder

80. Which of the following statements is incorrect?

- (1) Heat (q) has randomising influence on the system.
 (2) entropy change varies directly with temperature.
 (3) Entropy change is a state function
 (4) Temperature is the measure of average chaotic motion of the particle in the system.

81. The entropy change for
 $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow 3\text{H}_2(\text{g}) + \text{CO}(\text{g})$
 using the following data is

Substance	$\text{CH}_4(\text{g})$	$\text{H}_2\text{O}(\text{g})$	$\text{H}_2(\text{g})$	$\text{CO}(\text{g})$
$S^\circ/\text{JK}^{-1}\text{mol}^{-1}$	186.2	188.7	130.6	197.6

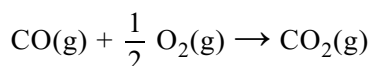
- (1) $-46 \text{ JK}^{-1} \text{ mol}^{-1}$
 (2) $+46 \text{ JK}^{-1} \text{ mol}^{-1}$
 (3) $-214.5 \text{ JK}^{-1} \text{ mol}^{-1}$
 (4) $214.5 \text{ JK}^{-1} \text{ mol}^{-1}$

82. For a reaction, $P+Q \rightarrow R+S$. The value of ΔH is -30 kJ mol^{-1} and ΔS is $-100 \text{ JK}^{-1} \text{ mol}^{-1}$. At what temperature the reaction will be at equilibrium?

- (1) 27°C (2) 52°C
 (3) 30°C (4) 45°C

83.

For the reaction,



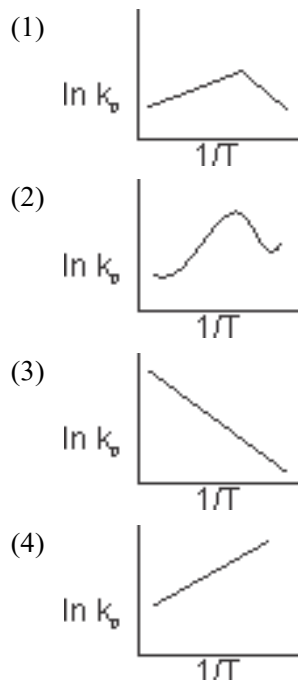
If the standard entropy change at 298 K is $0.082 \text{ kJ mol}^{-1} \text{ K}^{-1}$ and standard Gibb's energy for formation of CO_2 and CO are -298.4 and -114.6 kJ/mol respectively, then the enthalpy change for the given reaction will

- (1) -314.5 kJ/mol
 (2) -159.8 kJ/mol
 (3) -208.24 kJ/mol
 (4) -418.3 kJ/mol

84. The value of logarithm of equilibrium constant K_p if the standard free energy change of a reaction is $\Delta G^\circ = -115 \text{ kJ}$ at 298 K will be

- (1) 2.303
 (2) 13.83
 (3) 2.015
 (4) 20.15

85. Which of the following graphs represents exothermic reaction?



86. **Statement-I:** For every chemical reaction at equilibrium standard Gibbs energy of reaction is zero.

Statement-II: At constant temperature and pressure, chemical reactions are spontaneous in the direction of decreasing Gibbs energy.

- (1) Both statement I and statement II are incorrect
- (2) Statement I is correct and statement II is incorrect
- (3) Statement I is incorrect and statement II is correct
- (4) Both statement I and statement II are correct

87. Match the column I with Column II and mark the option given in the codes below.

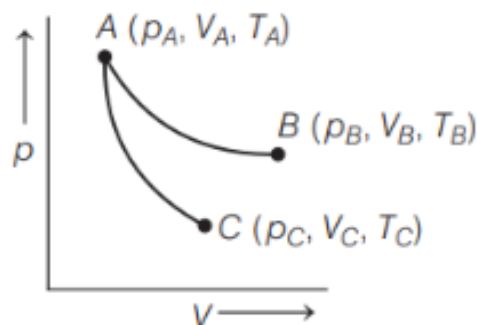
	Column -I		Column-II
A.	$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}$	1.	$\Delta_{\text{sol}} H^\circ$
B.	$\text{H}_2(\text{g}) \rightarrow 2\text{H}(\text{g})$	2.	$\Delta_{\text{lattice}} H^\circ$
C.	$\text{NaCl}(\text{s}) \rightarrow \text{Na}^+(\text{g}) + \text{Cl}^-(\text{g})$	3.	$\Delta_{\text{c}} H^\circ$
D.	$\text{NaCl}(\text{s}) \rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	4.	$\Delta_{\text{bond}} H^\circ$

- (1) (A) - 4, (B) - 3, (C) - 1, (D) - 2
- (2) (A) - 2, (B) - 1, (C) - 4, (D) - 3
- (3) (A) - 1, (B) - 2, (C) - 3, (D) - 4
- (4) (A) - 3, (B) - 4, (C) - 2, (D) - 1

88. In an adiabatic process, no transfer of heat takes place between system and surroundings. Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following

- (1) $q = 0$, $\Delta T \neq 0$, $W = 0$
- (2) $q \neq 0$, $\Delta T = 0$, $W = 0$
- (3) $q = 0$, $\Delta T = 0$, $W = 0$
- (4) $q = 0$, $\Delta T = 0$, $W \neq 0$

89. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure.



AB $\rightarrow$ Isothermal expansion

AC $\rightarrow$ Adiabatic expansion

Which of the following option is not correct?

- (1) $\Delta S_{\text{isothermal}} > \Delta S_{\text{adiabatic}}$
- (2) $T_A = T_B$
- (3) $W_{\text{isothermal}} > W_{\text{adiabatic}}$
- (4) $T_C > T_A$

90. Match List -I with List -II.

	List -I		List -II
A.	spontaneous process	1.	$\Delta H < 0$
B.	Process with $\Delta p = 0$, $\Delta T = 0$	2.	$\Delta G_{T,p} < 0$
C.	$\Delta H_{\text{reaction}}$	3.	Isothermal and isobaric process
D.	Exothermic process	4.	[Bond energies of molecules in reactants] - [Bond energies of product molecules]

Choose the correct answer from the options given below.

- (1) A - 3, B - 2, C - 4, D - 1
- (2) A - 2, B - 3, C - 4, D - 1
- (3) A - 2, B - 3, C - 1, D - 4
- (4) A - 2, B - 1, C - 3, D - 4

SUBJECT : BIOLOGY
Topic : GENETICS & EVOLUTION

91. In pedigree analysis, if both the parents are affected but, few progeny are unaffected then disorder is most likely to be
- (1) Autosomal recessive
 - (2) Autosomal dominant
 - (3) X-linked recessive
 - (4) Y-linked
92. Red eye female *Drosophila* is crossed with white eye male. If this cross result is 50% progeny being red eye, we can say that :-
- (1) Both parents were hemizygous
 - (2) female parent was heterozygous
 - (3) female parent was pure
 - (4) Both parents were pure
93. Study of chromosomal disorder is done through-
- (1) Pedigree analysis
 - (2) Cross hybridization experiment
 - (3) Karyotype analysis
 - (4) Reciprocal cross
94. Which is not always true for genes in diploid organisms ?
- (1) They occur in pairs
 - (2) They segregate at the time of gamete formation
 - (3) Only one of each pair is transmitted to a gamete
 - (4) One pair segregates independently of another pair

95. Read the following statements and answer the correct one :

Statement-I : VNTR belongs to a class of satellite DNA referred to as mini satellite.

Statement-II : VNTR varies in size from 0.1 to 20 kb.

- (1) Statement-I is true but statement-II is false
 - (2) Statement-I is false but statement-II is true
 - (3) Both statements are false
 - (4) Both statements are true
96. A dihybrid cross produces 6 phenotype in F_2 generation then which statement is correct
- (1) Both gene show dominance
 - (2) Both gene show incomplete dominance
 - (3) One is dominance and another is incomplete dominance
 - (4) Both gene show codominance
97. A dihybrid test cross produce following result :
- A – 1 : 1 : 1 : 1
- B – 1 : 1
- C – 7 : 1 : 1 : 7
- Which condition found in A, B and C

	A	B	C
(1)	Complete linkage	Independent assortment	Incomplete linkage
(2)	Independent assortment	Complete linkage	Incomplete linkage
(3)	Incomplete linkage	Independent assortment	Complete linkage
(4)	Incomplete linkage	Complete linkage	Independent assortment

98. Two genes are linked and are 12 map units apart. A heterozygous individual, whose parents are AAbb and aaBB, would be expected to produce gamete in the following frequencies :—

- (1) 44%-AB, 6%-Ab, 6%-aB, 44%-ab
- (2) 6%-AB, 44%-Ab, 44%-aB, 6%-ab
- (3) 12%-AB, 38%-Ab, 38%-aB, 12%-ab
- (4) 6%-AB, 6%-Ab, 44%-aB, 44%-ab

99. In which of the following inheritance phenotype of F_1 -hybrid does **not** resemble with either of its parents completely :-

- (1) Incomplete dominance
- (2) Codominance
- (3) Polygenic inheritance
- (4) All of above

100. In which of the following condition, both alleles will not be expressed together ?

- (1) $I^A I^B$
- (2) Aa
- (3) $Hb^A Hb^S$
- (4) Both 1 and 3

101.

	Column-A		Column-B
I.	Gregor Mendel	A.	Chromosomal theory of inheritance
II.	Sutton and Boveri	B.	Law of inheritance
III.	Henking	C.	Morgan
IV.	Linkage	D.	Discovered X-body

The correct match is :-

- (1) I-B, II-A, III-D, IV-C
- (2) I-D, II-B, III-A, IV-C
- (3) I-D, II-A, III-B, IV-C
- (4) I-B, II-C, III-D, IV-A

102. **Assertion** : Dominance is not an autonomous feature of a gene or its products.

Reason : Dominance depends on gene product and on the particular phenotype that we choose to examine, (when more than one phenotype is influenced by the same gene.)

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

103. Who observed that behaviour of chromosomes at meiosis can serve as the cellular basis of both segregation and independent assortment :-

- (1) Sutton & Boveri
- (2) Banden and Boveri
- (3) W. Fleming
- (4) Boveri and Brauer

104. When we perform selfing of heterozygous round (Bb) seed shape pea plant, then in their progeny, in what fraction do we obtain large size starch grain seed?

- (1) $\frac{1}{2}$
- (2) $\frac{3}{4}$
- (3) $\frac{1}{4}$
- (4) $\frac{2}{3}$

105. A person having codominant alleles for blood group marries a homozygous recessive woman, what will be the possible blood group of their offspring ?

- (1) AB or O
- (2) A or B
- (3) only AB
- (4) A, B or AB

- 106.** In a diploid organism how many maximum different alleles of a gene are present :-
 (1) One
 (2) Two
 (3) Many
 (4) Variable in different chromosome
- 107.** In the cross given below:
 $AaBbccDd \times AabbCCDd$
 What percentage of progeny will have genotype $aaBbCcDD$?
 (1) 6.2 % (2) 3.1 %
 (3) 12.4 % (4) 25 %
- 108. Assertion :-** Punnett square is a graphical representation to calculate the probability of all possible genotypes of offsprings in a genetic cross.
Reason :- It was developed by a British geneticist, Reginald C. Punnett.
 (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.
- 109.** Isotopes used for proving semiconservative replication of DNA are :-
 (1) N^{14} and P^{31} (2) N^{14} and N^{15}
 (3) N^{14} and C^{14} (4) S^{35} and P^{32}
- 110.** The sensitivities of DNA finger printing technique has been increased by use of-
 (1) Electrophoresis (2) PCR
 (3) Blotting (4) Hybridization
- 111.** Which of the following option is incorrect about transcription-
 (1) Both strands are not copied during transcription.
 (2) Main enzyme is DNA dependent RNA polymerase.
 (3) Primer synthesized by primase.
 (4) Promoter provides binding site for RNA polymerase.
- 112.** A segment of DNA has a base sequence.
 $ATG\ ACG\ GAC\ AGA\ AAC$
 Which one of the following represents frameshift mutation.
 (1) $GTC\ ATG\ AGA\ GAC\ AAC$
 (2) $ATG\ CAG\ GGA\ CGA\ AAC$
 (3) $ATG\ CGG\ ACA\ GAA\ AC$
 (4) $CAT\ GAC\ GAA\ CGA\ ACG$
- 113.** Which of the following is not associated with HGP :-
 (1) Expressed sequenced Tags (ESTs)
 (2) Sequence annotation
 (3) Automated DNA sequencer
 (4) Most commonly used vector is Ti-plasmid for cloning DNA segment
- 114.** In which one of the following combination (1-4) of number of chromosome in octaploid triticales (56) correctly represented :-

Combination	Monosomic	Haploid	Nullisomic	Trisomic
(1)	55	28	54	57
(2)	7	28	29	27
(3)	55	21	54	55
(4)	21	41	40	43

115. Which of the following is true for Lac-operon :-

- (i) Inactive repressor is formed
- (ii) Operates in anabolic pathway
- (iii) Inducer, inactivates repressor protein
- (iv) Three structural genes z,y,a are involved
- (v) i-gene synthesized repressor constitutively

- (1) i, ii, iii, iv
- (2) ii, iii, iv
- (3) ii, iii, v
- (4) iii, iv, v

116. Which of the following steps in transcription is catalysed by RNA polymerase alone ?

- (1) Initiation
- (2) Elongation
- (3) Termination
- (4) All of these

117. Enzyme that help in t-RNA formation in prokaryotes is :-

- (1) RNA polymerase-I
- (2) RNA polymerase-II
- (3) RNA polymerase-III
- (4) DNA dependent RNA polymerase

118. Read the following statements and choose the correct option.

Statement-I : Males generally function as carriers of sex linked disorders.

Statement-II : Mother passes the alleles of a sex linked trait to daughters only.

- (1) Both the statements are correct.
- (2) Only statement-I is correct
- (3) Only statement-II is correct
- (4) Both the statements are incorrect.

119. In "QB bacteriophage" genetic material is :-

- (1) DNA
- (2) RNA
- (3) Protein
- (4) Lipid

120. In eukaryotes RNA polymerase-I is responsible for transcription of :-

- (1) 28S r-RNA, 18S r-RNA, 5.8S r-RNA
- (2) hnRNA
- (3) t-RNA
- (4) Sn-RNA, 5S r-RNA

121. **Assertion** :- Inhibition of transcription is the only method of regulation of gene expression in all organisms.

Reason :- Translation is the first step of gene expression.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False
- (4) Both Assertion & Reason are false.

122. Which of the following is not a part of transcription unit ?

- (1) Promoter
- (2) Structural gene
- (3) Terminator
- (4) Initiator codon

123. In full turn of B-DNA. How many N_2 -bases are present :-

- (1) 10
- (2) 20
- (3) 30
- (4) 5

124. On the DNA template with polarity 3'-5', the replication is:

- (1) Continuous
- (2) Discontinuous
- (3) Both (1) and (2)
- (4) Uncertain

125. A N^{15} DNA molecule is replicated in N^{14} medium & obtained 4th generation. If it is centrifused just after denaturation than there will be :-

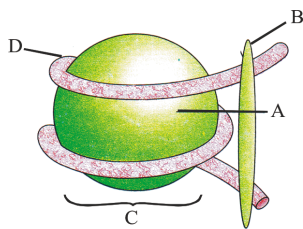
- (1) No band
- (2) Two band, one of light DNA & 2nd of heavy DNA
- (3) Two band one of light DNA & one of hybrid DNA
- (4) Three band each for light, heavy & hybrid.

126. **Statement-I** : In honey bee males produce sperm by mitosis and unfertilized egg develops as a drone.

Statement-II : In grasshopper males have only one X-chromosome besides the autosomes.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II are incorrect.

127. Choose the correct match :-



- (1) A-Histone octamer, B-DNA, C- Core of histone molecule, D- H_1 Histone
- (2) A-DNA, B- H_1 Histone, C- Core of histone molecule, D-Histone octamer
- (3) A - Histone octamer, B - H_1 Histone, C - Core of histone molecule, D - DNA
- (4) A - Core of histone molecule, B - DNA, C - Histone octamer, D - H_1 Histone

128. Deoxythymidine contains :-

- (a) Thymine
- (b) Ribose
- (c) Deoxyribose
- (d) Purine base
- (e) Pyrimidine base

- (1) a, b and e
- (2) a, c and d
- (3) a, b and d
- (4) a, c and e

129. Who proposed the semiconservative mode of DNA replication:

- (1) Taylor
- (2) Severo ochoa
- (3) Watson and crick
- (4) Gamow

130. **Statement-I** : Leaf colour in *Mirabilis jalapa* is an example of incomplete dominance.

Statement-II : F_1 resembles with any one of the parents in case of incomplete dominance.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II are incorrect.

131.

	Organism		No. of nucleotides
(i)	$\phi \times 174$ bacteriophage	(A)	4.6×10^6 bp
(ii)	Lambda bacteriophage	(B)	6.6×10^9 bp
(iii)	<i>E.Coli</i>	(C)	5386 nucleotides
(iv)	Human	(D)	48502 bp

Following match is correct :-

- (1) i = D, ii = C, iii = B, iv = A
- (2) i = B, ii = D, iii = A, iv = B
- (3) i = A, ii = D, iii = C, iv = B
- (4) i = C, ii = D, iii = A, iv = B

132. **Statement I** : In Eukaryotes, transcription occurs in nucleus.

Statement II : In Bacteria, transcription and translation occur in cytoplasm.

- (1) Both statement I and II are true
- (2) Both statement I and II are false
- (3) Statement I is true but II is false
- (4) Statement I is false but II is true

133. Match column-I with column-II and select the correct option from the given codes.

	Column-I		Column-II
A	Sigma factor	i	5'-3'
B	Capping	ii	Initiation
C	Tailing	iii	Termination
D	Coding strand	iv	5' end
		v	3' end

- (1) A-iii, B-iv, C-iv, D-ii (2) A-ii, B-iv, C-v, D-i
- (3) A-ii, B-iv, C-v, D-iii (4) A-iii, B-v, C-iv, D-i

134. In Oswald Avery, Colin MacLeod and Maclyn McCarty experiment transforming substance was digested by using :-

- (1) DNase (2) RNase
- (3) Proteases (4) High temperature

135. Match the column-I with column-II and select the correct answer :-

	Column-I		Column-II
(A)	Monohybrid cross	(i)	TtRr
(B)	Test cross	(ii)	Tt × TT
(C)	Out cross	(iii)	Tt × tt
(D)	True dihybrid	(iv)	Tt × Tt
(E)	Monohybrid	(v)	Rr

- (1) A – iv, B – iii, C – ii, D – i, E – v
- (2) A – i, B – ii, C – iii, D – iv, E – v
- (3) A – iii, B – i, C – ii, D – iv, E – v
- (4) A – ii, B – iv, C – iii, D – i, E – v

136. Which is not true about fossils ?

- (1) Fossils are always the hard parts of past organisms
- (2) Mostly fossils are found in sedimentary rocks
- (3) Different aged rock sediments contain fossils of different life forms
- (4) The age of a fossil can be estimated by radioactive dating

137. Which organisms probably evolved into first amphibians?

- (1) Synapsids
- (2) Sauropsids
- (3) Coelacanth
- (4) Ichthyosaurs

138. Which is not true for homologous structures or homology?

- (1) Homologous structures are same in origin
- (2) Homologous structures always perform same function
- (3) Homology is based on divergent evolution
- (4) Homology indicates common ancestry

139. S.L. Miller an American scientist created electric discharge in a closed flask containing ____ (A) ____ at ____ (B) ____.

- (1) A - CH₄, H₂, NH₃ and H₂O vapour, B - 800°C
- (2) A - CH₄, CO₂ and H₂O vapour, B - 800°C
- (3) A - CH₄, CO₂, NH₃ and H₂O vapour, B - 600°C
- (4) A - CH₄, O₂, NH₃, H₂ and H₂O vapour, B - 600°C

140. Which of the following is the correct order for evolution of mammals?

- (1) Early reptiles → Sauropsids → Synapsids → Mammals
- (2) Synapsids → Pelycosaurs → Thecodonts → Mammals
- (3) Sauropsids → Pelycosaurs → Therapsids → Mammals
- (4) Synapsids → Pelycosaurs → Therapsids → Mammals

141. Louis Pasteur disproved the theory of spontaneous generation.

Which of the following helped him in doing so?

- (1) Living organisms were formed in a closed pre-sterilised flask
- (2) Living organisms were found in decaying and rotting organic matter
- (3) New living organisms were found only in a flask whose neck was broken
- (4) Electric discharge in a closed flask with water vapour and other gases

142. Which one of the following options gives one correct example each of convergent evolution and divergent evolution :-

	Convergent evolution	Divergent evolution
(1)	Eyes of octopus and mammals	Bones of fore limbs of vertebrates
(2)	Thorns of Bougainvillea and tendrils of cucurbita	Wings of butterflies and birds
(3)	Bones of fore limbs of vertebrates	Wings of butterfly and birds
(4)	Thorns of Bougainvillea and tendrils of cucurbita	Eyes of octopus and mammals

143. Assertion: During evolution, pouched mammals of Australia survived.

Reason: Due to lack of competition from any other mammal due to continental drift.

- (1) Both Assertion & Reason are true, and the reason is the correct explanation of the assertion.
- (2) Both Assertion & Reason are true, but the reason is not the correct explanation of the assertion.
- (3) Assertion is a true statement, but Reason is false.
- (4) Both Assertion and Reason are false statements.

144. Mutations areA..... andB..... while Darwinian variations areC..... andD.... Fill in the blanks for A,B,C and D from the options given below-

- (1) A- Random, B- Directional, C- Small, D – Directionless
- (2) A- Small, B- Directional, C- Random, D – Directionless
- (3) A- Random, B- Directionless, C- Small, D – Directional
- (4) A- Small, B- Directionless, C- Random, D – Directional

145. Adaptive radiation that have occurred in placental mammals and Australian marsupials in isolated geographical areas resulted in convergent evolution, Select correct pair of placental mammal (Column-I) and marsupial (Column-II) exhibiting convergent evolution:-

	Column-I	Column-II
(1)		
(2)		
(3)		
(4)		

146. Darwin's finches are good example of
(A) Divergent evolution
(B) Allopatric speciation
(C) Adaptive radiation
(1) A and C (2) C only
(3) B only (4) A, B and C
147. Which of the following had the smallest brain capacity
(1) Homo sapiens
(2) Homo neanderthalensis
(3) Homo habilis
(4) Homo erectus
148. Panspermia, an idea that is still a favourite for some astronomers, means :
(1) Creation of life from dead and decaying matter
(2) Creation of life from chemicals
(3) Origin of sperm in human testes
(4) Transfer of spores as unit of life from other planets of Earth
149. Genetic Drift occurs due to ?
(1) Sudden population migration
(2) Continuous gene migration
(3) Mutation
(4) Natural selection
150. Out of the following how many statements are correct?
(A) Fitness is based on characteristics which are inherited.
(B) Fitness is the end result of ability to adapt & get selected by nature.
(C) Adaptive ability is inherited
(D) Natural selection is based on certain observations which are not factual.
(1) A & C only (2) A, B & C
(3) A & D only (4) A & B only
151. Which of the following is incorrect ?
(1) Earth originated about 4.5 billion years back
(2) The first organisms were chemoautotroph
(3) Experimental proof that some simple molecules like H_2 , NH_3 , CH_4 and H_2O gave rise to amino acids during origin of life was provided by S.L. Miller
(4) Oxygen was absent at the time of origin of earth
152. According to the theory of Special creation :
(1) All living organisms as we see today were created as such
(2) Earth is about 4000 years old
(3) Diversity was always same since creation and will be the same in future
(4) All the above three options are true
153. Alfred Wallace, a naturalist:
(1) Worked in Malay Archipelago
(2) Had conclusion similar to those of Charles Darwin
(3) Believed in fixity of species
(4) Both (1) and (2)
154. If 64% of the individuals in a population are homozygous recessive for a character then calculate the percentage of homozygous dominant individuals in this population :
(1) 4% (2) 0.4% (3) 0.8% (4) 32%
155. Identify the correct match w.r.t convergent evolution:
(1) Anteater — Numbat
(2) Bobcat — Spotted cuscus
(3) Anteater — Spotted cuscus
(4) Tasmanian tiger cat — Wolf

- 156.** The correct sequence of stage in evolution of modern man (*Homo sapiens sapiens*) is :
- Homo erectus*
 - Neanderthal man
 - Modern man
 - Australopithecus*
 - Cro-Magnon man
- IV $\rightarrow$ II $\rightarrow$ I $\rightarrow$ III $\rightarrow$ I
 - IV $\rightarrow$ I $\rightarrow$ II $\rightarrow$ V $\rightarrow$ III
 - IV $\rightarrow$ II $\rightarrow$ III $\rightarrow$ I $\rightarrow$ V
 - IV $\rightarrow$ V $\rightarrow$ III $\rightarrow$ I $\rightarrow$ II
- 157.** Darwin didn't believe in:
- Unchangeable fauna and flora on earth
 - Descent with modification
 - Reproductive fitness of organisms
 - In-built variations among the members of a populations
- 158.** The probable first prehistoric man was :
- Ramapithecus*
 - Homo habilis*
 - Australopithecus*
 - Sinanthropus*
- 159.** New breeds in domestic dogs are developed by :
- Sexual selection
 - Natural selection
 - Youth selection
 - Artificial selection
- 160.** Consider the following four statements (A-D) and select the option which includes all the correct ones only -
- The Neanderthal man had a brain size of 1400 cc
 - Homo sapiens* man arose in America
 - Agriculture came around 10000 years back
 - Neanderthal man used hides to protect their body and buried their dead
- A, B and C
 - A, B and D
 - A, C and D
 - B, C and D
- 161.** The ancestors of human who hunted with stone weapons but essentially ate fruits, were ?
- Ramapithecus*
 - Dryopithecus*
 - Australopithecus*
 - Homo erectus*
- 162. Assertion :** When we see stars, we apparently are peeping into the past.
Reason : The light emitted from stars started its journey millions of years back and from trillions of kilometers away and reaching our eyes now.
- Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 - Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 - Assertion is True but the Reason is False.
 - Both Assertion & Reason are False.
- 163.** Life cannot originate from inorganic materials at present because—
- High degree of environmental pollution
 - A very high amount of oxygen in the atmosphere
 - Very high atmospheric temperature
 - Absence of raw materials
- 164. Assertion :-** Lichens can be used as industrial pollution indicator.
Reason :- Lichens forces the moth to change as dark winged or white winged in industrial area.
- Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 - Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 - Assertion is True but the Reason is False.
 - Both Assertion & Reason are False.

165. Select the incorrect statement :-

- (1) Universe is very old-almost 20 billion year old.
- (2) Considering the size of earth, universe is indeed a speck.
- (3) There was no atmosphere on early earth.
- (4) Earth is found in milky way galaxy.

166. Scale From Oldest to Youngest :

a – Carboniferous, b – Devonian
c – Permian, d – Silurian

- (1) $d \rightarrow b \rightarrow c \rightarrow a$
- (2) $d \rightarrow b \rightarrow a \rightarrow c$
- (3) $c \rightarrow a \rightarrow d \rightarrow b$
- (4) $c \rightarrow d \rightarrow b \rightarrow a$

167. Which of the following is the **correct** sequence of events in the origin of life ?

- (A) Formation of protobionts
- (B) Synthesis of organic monomers
- (C) Synthesis of organic polymers
- (D) Formation of prokaryote

- (1) B, C, A, D
- (2) B, C, D, A
- (3) A, B, C, D
- (4) A, C, B, D

168. Charles Darwin concluded after his sea voyage

- (1) Existing living forms share similarities to varying degrees among themselves
- (2) Existing living forms share similarities to life forms that existed millions of years ago
- (3) Existing living forms are totally dissimilar with life forms that existed millions of years ago
- (4) Both (1) and (2)

169. Given below are two statements :

Statement-I :- The first organisms that invaded land were plants.

Statement-II :- First cellular form life did not possibly originate before about 2000 million years ago.

- (1) Both Statements I and Statement II are false
- (2) Statement I is true but Statement II is false
- (3) Statement I is false but Statement II is true
- (4) Both Statement I and Statement II are true

170. Which statement is wrong for Darwin's theory

- (1) All population has built in variation in Characteristics
- (2) Organism whose characteristics which are better for survival in natural conditions breed less and survive
- (3) Those who are better fit in environment survive more and naturally selected
- (4) Natural selection was taken as a mechanism of evolution

171. Which of the following is an extinct organism

- (1) Whale
- (2) Dinosaurs
- (3) Tiger
- (4) Rhinoceros

172. Before industrialization in England which variety of moths were found abundantly

- (1) White winged
- (2) Dark color winged
- (3) Moths with one white and other dark colour moth
- (4) Moths with small wings

173. Neo darwinism believes that new species develop through :-

- (1) Mutation
- (2) Hybridization
- (3) Mutation with natural selection
- (4) Production

174. **Assertion** :- Evolution is not a directed process in the sense of determinism.

Reason :- Evolution is a stochastic process based on chance events in nature and chance mutation in the organisms.

- (1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
- (2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion and Reason are False.

175. "Embryo never pass through the adult stages of any animal" concluded by -

- (1) Von-Baer
- (2) Haeckel
- (3) Muller
- (4) Darwin

176. Choose the wrong statement regarding Hardy-Weinberg principle?

- (1) Allele frequencies in population may be stable and constant from generation to generation
- (2) Variation due to genetic drift results in changed frequency of genes and alleles in future generations
- (3) Genetic recombination helps in maintaining Hardy-Weinberg equilibrium
- (4) Mutations can change Hardy weinberg equilibrium

177. The changed frequency of genes and alleles in future generation occur through-

- (1) Mutation
- (2) Variation due to recombination
- (3) Gene flow or genetic drift
- (4) All of these

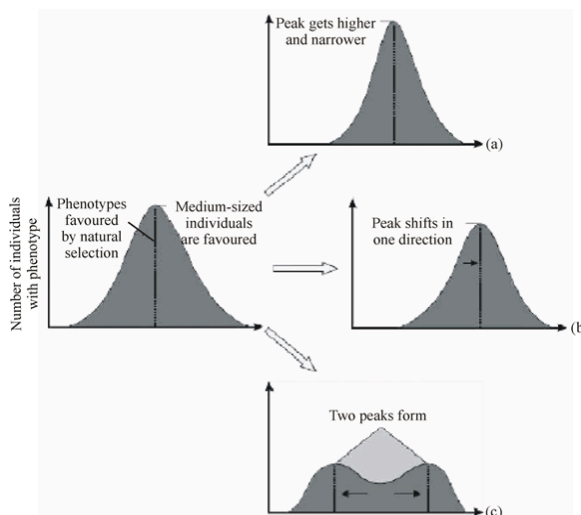
178. The major defect of darwinism was:

- (1) Non-description of survival of fittest
- (2) Non-description of reason for variation
- (3) Non - description of natural selection
- (4) Non description of over-production of young ones

179. Example of fish like reptile is :

- (1) Ichthyosaurs
- (2) Tyranosaurs
- (3) Triceratops
- (4) Crocodile

180. Recognise the figure and find out the correct matching.



- (1) b-directional, a-disruptive, c-stabilising
- (2) b-directional, c-disruptive, a-stabilising
- (3) c-directional, b-disruptive, a-stabilising
- (4) a-directional, c-disruptive, b-stabilising

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